

FEDERAL PROJECT NO X112-221-L1CE; X112-221-L680																				
BOO11-0	DISTRICT	COUNTY					TOWNSHIP					BOROUGH			ROUTE	SECTION	TOTAL SHEETS			
	11-0	BEAVER					ROCHESTER					FALLSTON			0018	B35				
												BRIDGEWATER			0051	B30				
												FALLSTON			0051	B30				
WBS ELEMENT																				
T/P		SYS	WO					SPUR	PHA	SECTION			ORG.			PRG.			P.C	
P		0	9	6	3	7	1	0	7	0	0	0	1	1	2	0	2	2	3	1
P		0	9	6	3	7	1	0	7	0	0	0	1	1	2	0	2	2	3	2
P		0	9	6	3	7	1	0	7	0	0	0	1	1	2	0	3	6	5	1

ECMS NO 28898

COMMONWEALTH OF PENNSYLVANIA



DEPARTMENT OF TRANSPORTATION

DRAWINGS FOR CONSTRUCTION

OF VETERANS MEMORIAL BRIDGE (COUNTY BRIDGE NO 63) AND APPROACHES IN BEAVER COUNTY

FROM STA 10+25.65 TO STA 19+64.04 LENGTH 938.39 FT 0.178 MI

AND

STATE ROUTE 0051 SECTION B30

FROM STA 829+00.00 TO STA 874+15.12 LENGTH 4411.81 FT 0.836 MI

SEGMENT 0260 OFFSET 0945 TO SEGMENT 0240 OFFSET 1812

SEGMENT 0261 OFFSET 0945 TO SEGMENT 0241 OFFSET 1821

AND

STATE ROUTE 0018 SECTION B35

FROM STA 48+35.63 TO STA 81+16.89 LENGTH 3181.26 FT 0.602 MI

SEGMENT 0550 OFFSET 1217 TO SEGMENT 0560 OFFSET 1566

SEGMENT 0551 OFFSET 1153 TO SEGMENT 0561 OFFSET 1566

TOTAL LENGTH 8770.77 FT 1.661 MI

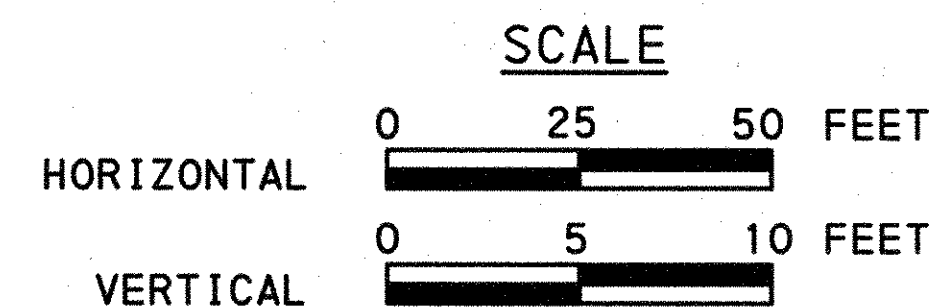
ALSO INCLUDED:

TRAFFIC CONTROL PLAN	30 SHEETS
SIGNING, PAVEMENT MARKING AND DELINEATION PLAN	20 SHEETS
EROSION AND SEDIMENT POLLUTION CONTROL PLAN	16 SHEETS
LIGHTING PLAN	10 SHEETS
RAILROAD RELOCATION PLAN	63 SHEETS
TRAFFIC SIGNAL PLAN	20 SHEETS
SHARON ROAD WATERLINE	3 SHEETS
STRUCTURE PLANS	
BPAA 04-2673 (RAILROAD BRIDGE)	77 SHEETS
BPAA 04-2719 (RIVER BRIDGE)	106 SHEETS
CROSS SECTIONS	39 SHEETS
CROSS SECTIONS (RAILROAD RELOCATION)	57 SHEETS
CONCEPTUAL WALL DESIGN	2 SHEETS
EXISTING BRIDGE PLANS	6 SHEETS
BR #27.34	

SR 0051 PREVIOUSLY KNOWN AS LR 204
SR 0018 PREVIOUSLY KNOWN AS LR 77
SR 0065 PREVIOUSLY KNOWN AS LR 641

SR 0051 ESTABLISHED AS A LIMITED ACCESS HIGHWAY FROM
STATION 848+16.50 TO STATION 853+80.00 BY PLAN OF
ROUTE NO 204, SECTION NO 11 RIGHT-OF-WAY PLAN
APPROVED FEBRUARY 2, 1956.

SR 0018/0065 ESTABLISHED AS A LIMITED ACCESS HIGHWAY
FROM STATION 48+85.63 TO 80+66.89 BY PLAN OF ROUTE
NO 77 SECTION NO 16 AND ROUTE NO 641 SECTION NO 20
RIGHT-OF-WAY PLAN APPROVED JANUARY 7, 1958.



VETERANS MEMORIAL BRIDGE DESIGN DESIGNATION

HIGHWAY CLASSIFICATION - URBAN ARTERIAL
DESIGN SPEED - 40 MPH
PAVEMENT WIDTH - 3 - 12 FT LANES
SHOULDER WIDTH - 8'
MEDIAN WIDTH - N/A

TRAFFIC DATA

CURRENT YEAR ADT - 7,070 (2010)
DESIGN YEAR ADT - 10,300 (2030)
DHV - 1,030
D - 50%
T - 10%

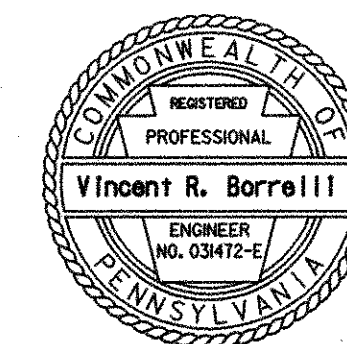
PUC APPLICATION DOCKET NUMBER A-00121999

PLANS PREPARED BY

MICHAEL BAKER JR., INC.
Consulting Engineers
AIRSIDE BUSINESS PARK
100 AIRSIDE DRIVE
MOON TOWNSHIP, PA 15108

Baker

MICHAEL BAKER JR., INC.



Vincent R. Borrelli 6/16/10
PROJECT MANAGER DATE
REGISTERED PROFESSIONAL ENGINEER

RECOMMENDED JUNE 17, 2010

Steve Castor
DISTRICT EXECUTIVE

RECOMMENDED JUNE 16, 2010

Tommy Amadio
BEAVER COUNTY COMMISSIONER

RECOMMENDED JUNE 16, 2010

Joe Spauld
BEAVER COUNTY COMMISSIONER

RECOMMENDED JUNE 16, 2010

Charles R. Camp
BEAVER COUNTY COMMISSIONER

TABULATION OF SEGMENT EQUALITIES

SR 0051					
SEGMENT 0260	OFFSET 0000 NB	=	STA 838+45.00		
SEGMENT 0261	OFFSET 0000 SB	=	STA 838+45.00		
SEGMENT 0250	OFFSET 0000 NB	=	STA 871+88.00		
SEGMENT 0251	OFFSET 0000 SB	=	STA 872+83.00		
SEGMENT 0240	OFFSET 0000 NB	=	STA 892+27.00		
SEGMENT 0241	OFFSET 0000 SB	=	STA 892+36.00		
SR 0018					
SEGMENT 0560	OFFSET 0000 NB	=	STA 65+51.00		
SEGMENT 0561	OFFSET 0000 SB	=	STA 65+51.00		
SEGMENT 0550	OFFSET 0000 NB	=	STA 36+18.00		
SEGMENT 0551	OFFSET 0000 SB	=	STA 36+18.00		

LEGEND

- △ PARCEL IDENTIFICATION NUMBER - NO TAKE
- PARCEL IDENTIFICATION NUMBER
- ⊙ PLAN
- PROFILE
- SHEET LIMITS

STOP WORK
STA 8+56.39
SHARON ROAD

START WORK
STA 5+42.69
SHARON ROAD

POT STA 5+02.45 SHARON ROAD =
POT STA 65+48.40 SR 0018

POT STA 19+90.20 VETERANS MEMORIAL BRIDGE =
POT STA 65+44.82 SR 0018

POT STA 18+59.04 VETERANS MEMORIAL BRIDGE =
POC STA 2026+73.22 PERMANENT TRACK 1 C

POT STA 18+45.00 VETERANS MEMORIAL BRIDGE =
POC STA 4026+71.45 PERMANENT TRACK 2 C

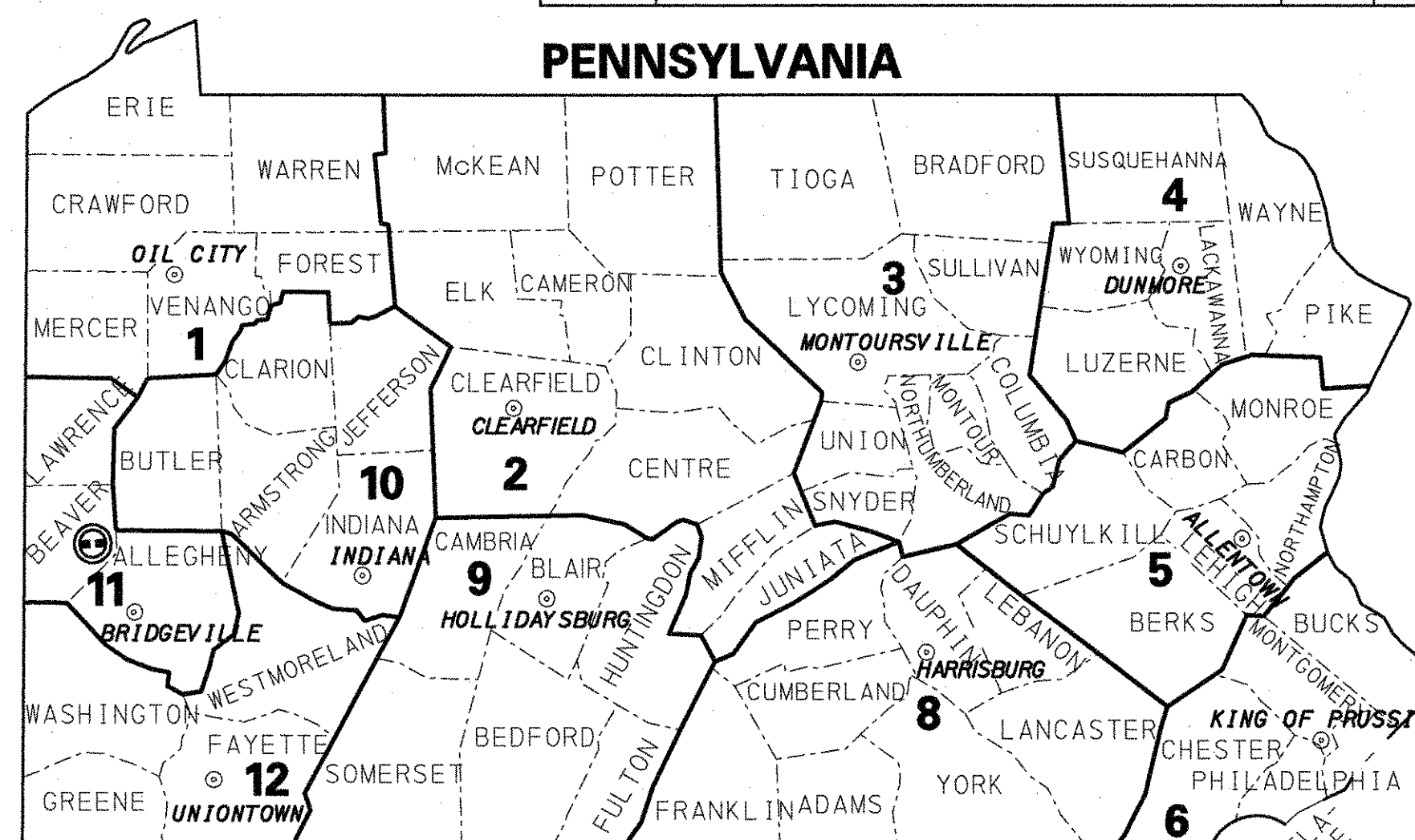
POT STA 18+30.95 VETERANS MEMORIAL BRIDGE =
POC STA 6026+69.26 PERMANENT TRACK 3 C

POT STA 18+16.91 VETERANS MEMORIAL BRIDGE =
POC STA 8026+67.36 PERMANENT TRACK 4 C

PERMANENT TRACK 4 CONSTR C
PERMANENT TRACK 3 CONSTR C
PERMANENT TRACK 2 CONSTR C
PERMANENT TRACK 1 SURVEY & CONSTR C

STA 18+37.56 VETERANS MEMORIAL BRIDGE =
STA 11+62.50 RAILROAD BRIDGE
STOP WORK
STA 80+66.89
SEGMENT 0560 OFFSET 1516
SEGMENT 0561 OFFSET 1516
SR 0018 SECTION B35

LIMIT OF WORK
STA 81+16.89
SEGMENT 0560 OFFSET 1566
SEGMENT 0561 OFFSET 1566
SR 0018 SECTION B35
ROCHESTER TOWNSHIP
BEAVER COUNTY



11 DISTRICT
DISTRICT OFFICE
PROJECT LOCATION

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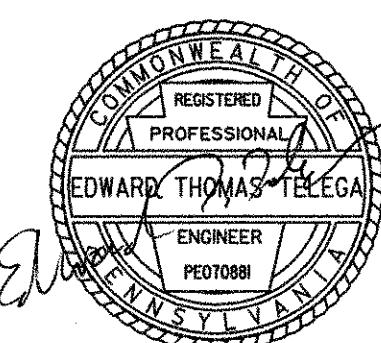
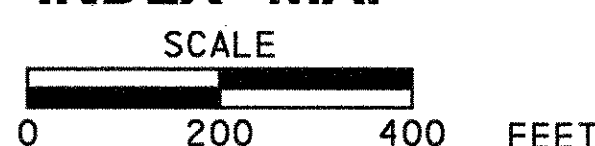
LIMIT OF WORK
STA 874+15.12
SEGMENT 0240 OFFSET 1812
SEGMENT 0241 OFFSET 1821
SR 0051 SECTION B30
BOROUGH OF BRIDGEWATER
BEAVER COUNTY

STOP WORK
STA 873+65.12
SEGMENT 0240 OFFSET 1862
SEGMENT 0241 OFFSET 1871
SR 0051 SECTION B30

PROPERTY OWNERS

- 1 WRIGHT CHIPPEWA HOLDINGS, LP, A LIMITED PARTNERSHIP
- 2 CSX TRANSPORTATION, INC.
- 3 BEAVER RIVER ENTERPRISES, LP
- 4 WRIGHT CHIPPEWA HOLDINGS, LP, A LIMITED PARTNERSHIP
- 5 DAVID KANE AND SARAH KANE, HUSBAND AND WIFE
- 6 KEN JARAM
- 7 DUQUESNE LIGHT COMPANY
- 8 DAVID FRANK KANE & SARAH M. KANE, HUSBAND AND WIFE
- 9 DAVID KANE AND SARAH KANE, HUSBAND AND WIFE
- 10 DAVID F. KANE AND SARAH M. KANE, HUSBAND AND WIFE
- 11 BOVARD-ANDERSON COMPANY, A PENNSYLVANIA CORPORATION
- 12 ALBERT J. SEBASTIAN AND JULIA M. SEBASTIAN, HIS WIFE
- 13 DAVID F. KANE & SARAH M. KANE, HUSBAND AND WIFE
- 14 UNKNOWN
- 15 BRUCE REED
- 16 CINDLEE CORP., AN OHIO CORPORATION
- 17 BEAVER COUNTY COMMUNITY BUILDING COMPANY
- 18 VOLLMER MOTORS SALES, INC.
- 19 NORFOLK SOUTHERN RAILWAY COMPANY
- 20 NEW BRIGHTON BOROUGH SANITARY AUTHORITY
- 21 JOHN MEDVED, MARRIED MAN
- 22 RWK REAL ESTATE, A PENNSYLVANIA PARTNERSHIP
- 23 UNITED REFINING COMPANY OF PENNSYLVANIA
- 27 BEAVER VALLEY TRANSIT COMPANY

INDEX MAP

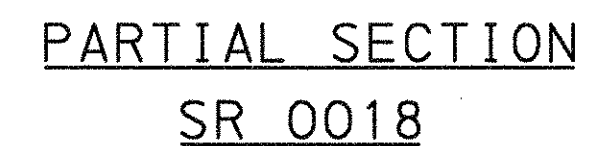
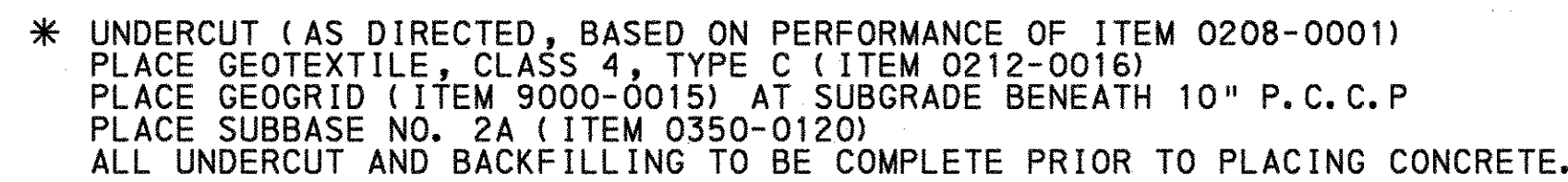
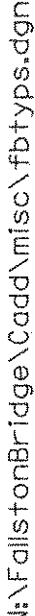


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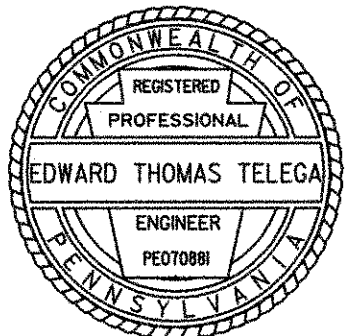
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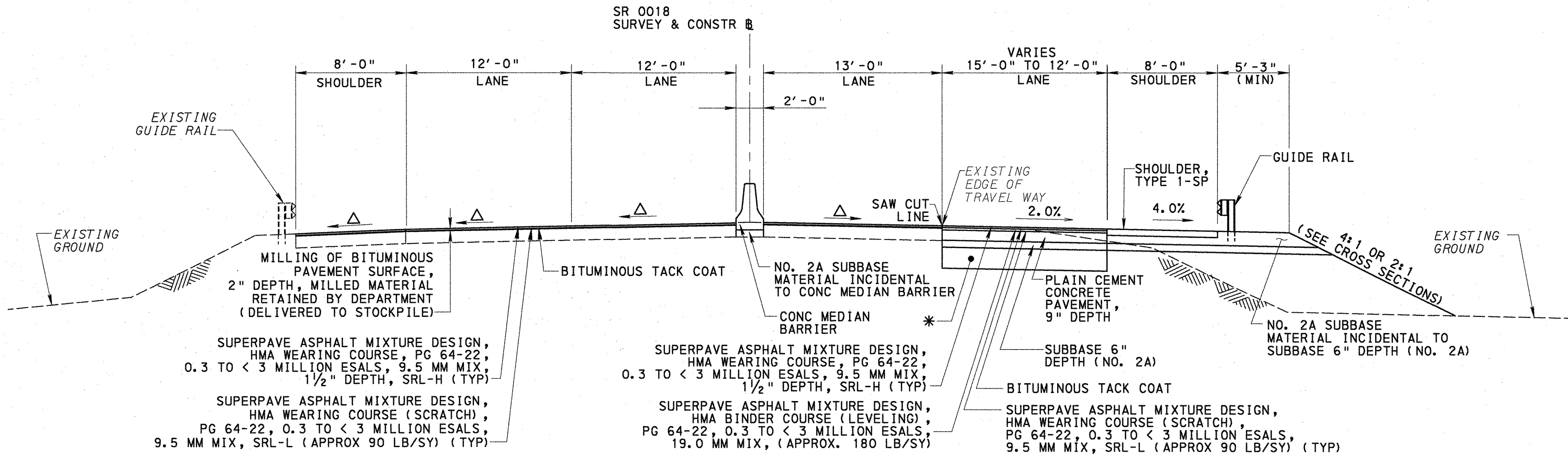
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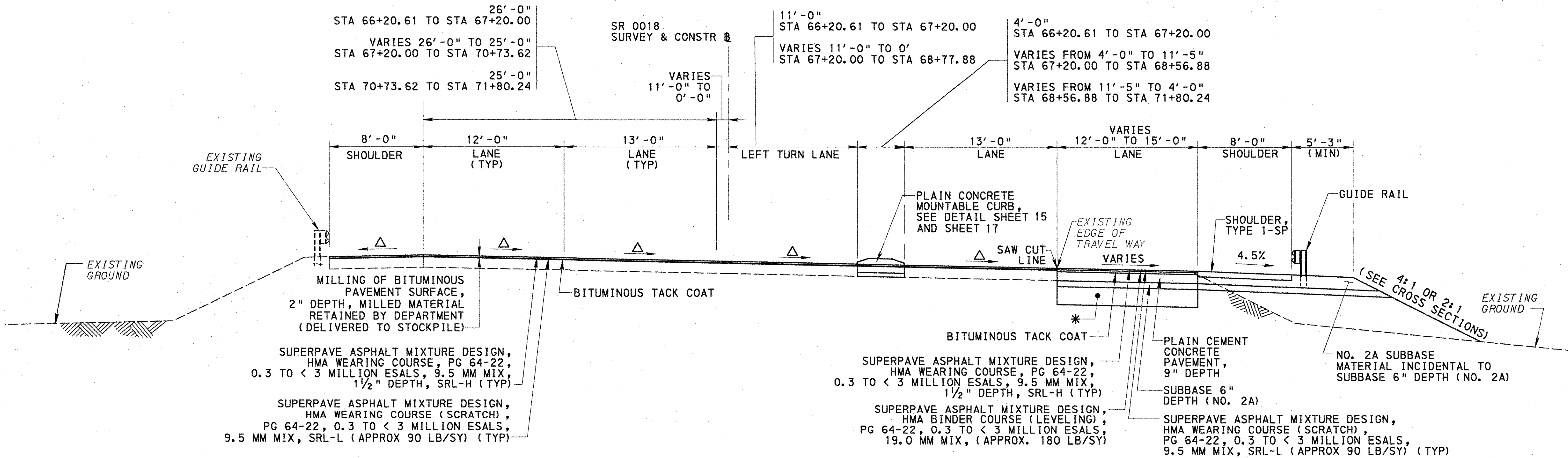
SEED SLOPES STEEPER THAN 3:1 WITH FORMULA L SEED MIXTURE.
SEED SLOPES 3:1 AND FLATTER WITH FORMULA D SEED MIXTURE.



DISTRICT	COUNTY	ROUTE	SECTION	SHEET
11-0	BEAVER	0051	B30	11 OF 68
		0018	B35	
FALLSTON BOROUGH, BOROUGH OF BRIDGEWATER, ROCHESTER TOWNSHIP				
REVISION NUMBER	REVISIONS			DATE BY



TYPICAL TANGENT SECTION
SR 0018
STA 71+80.24 TO STA 73+00.00
NOT TO SCALE



TYPICAL SUPERELEVATED SECTION
SR 0018
STA 66+20.61 TO STA 71+80.24
NOT TO SCALE

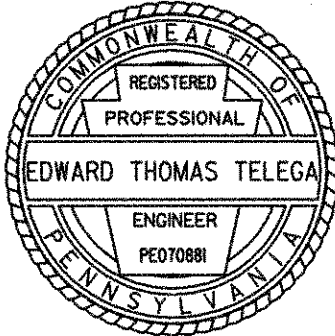
* UNDERCUT (AS DIRECTED, BASED ON PERFORMANCE OF ITEM 0208-0001)
PLACE GEOTEXTILE, CLASS 4, TYPE C (ITEM 0212-0016)
PLACE GEOGRID (ITEM 9000-0015) AT SUBGRADE BENEATH 10" P.C.C.P
PLACE SUBBASE NO. 2A (ITEM 0350-0120)
ALL UNDERCUT AND BACKFILLING TO BE COMPLETE PRIOR TO PLACING CONCRETE.

NOTE:
APPLY BITUMINOUS TACK COAT BEFORE
PLACING EACH BITUMINOUS LAYER, ON
EXISTING AND NEW SURFACES, INCLUDING
MULTIPLE LIFTS OF THE SAME LAYER.

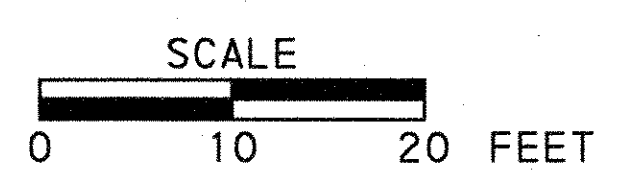
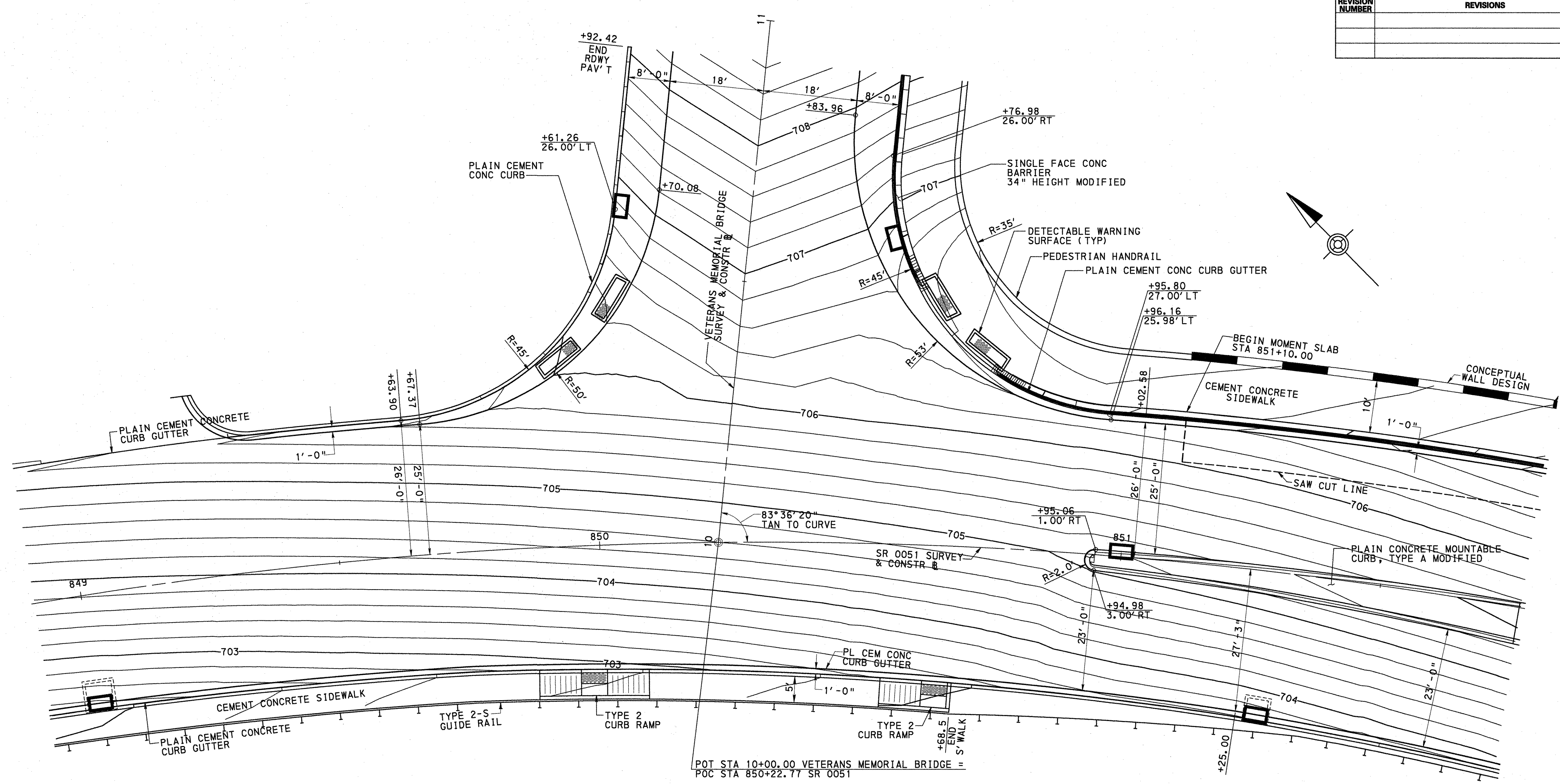
SEEDING LEGEND

SEED SLOPES STEEPER THAN 3:1 WITH FORMULA L SEED MIXTURE.
SEED SLOPES 3:1 AND FLATTER WITH FORMULA D SEED MIXTURE.

Δ - MATCH EXISTING PAVEMENT SLOPE

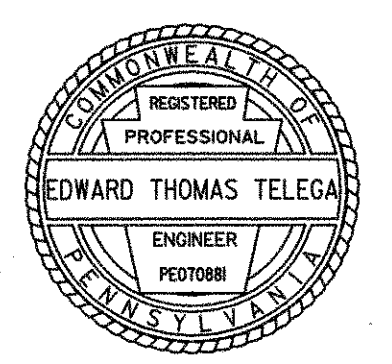


DISTRICT	COUNTY	ROUTE	SECTION	SHEET	
11-0	BEAVER	0051	B30	12 OF 68	
		0018	B35		
BOROUGH OF BRIDGEWATER					
REVISION NUMBER	REVISIONS			DATE	BY



CONTOUR INTERVAL = 0.2'

**INTERSECTION DETAIL
SR 0051 AND
VETERANS MEMORIAL BRIDGE**

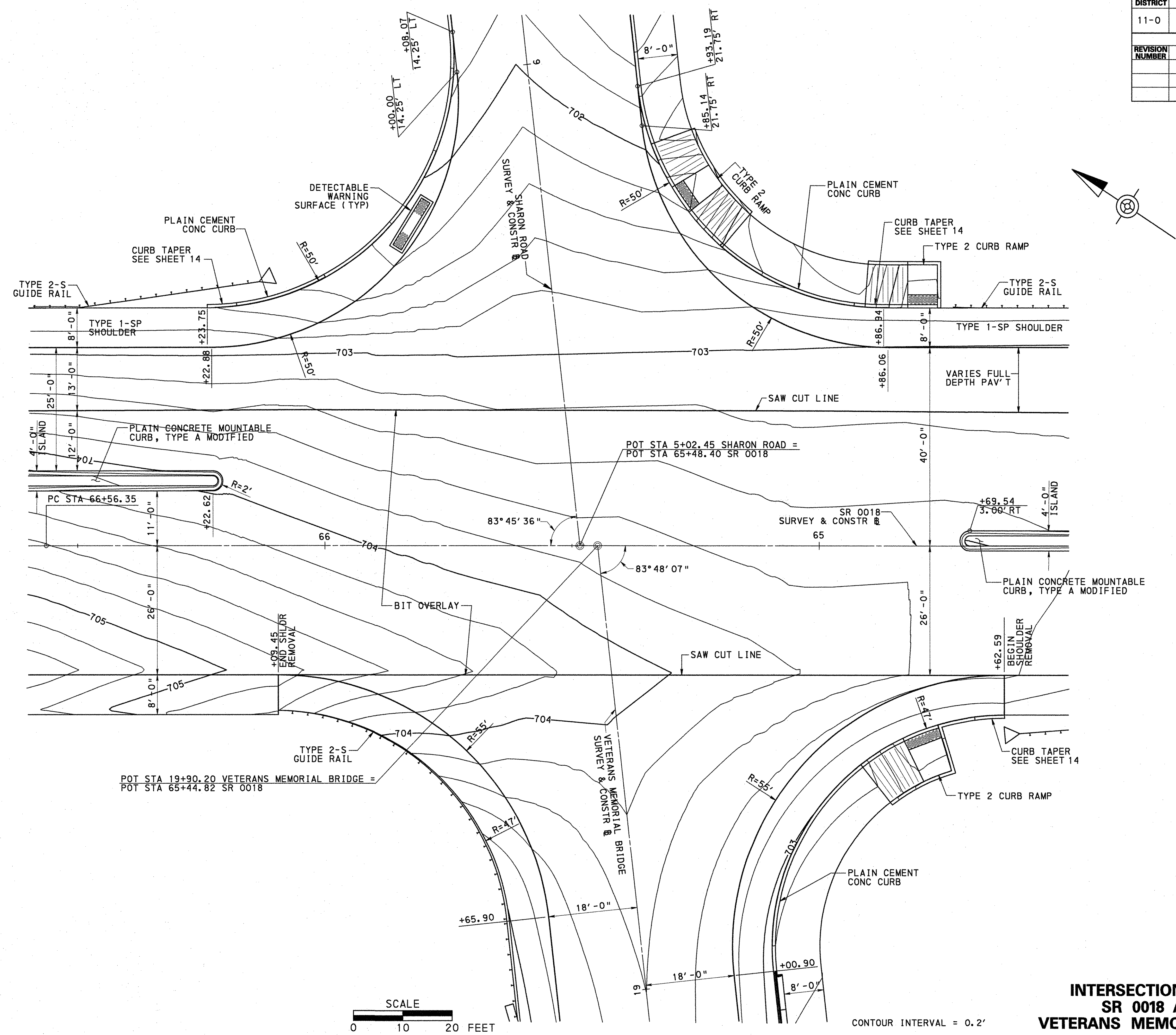


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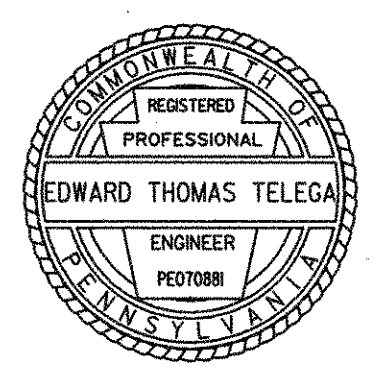
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DISTRICT	COUNTY	ROUTE	SECTION	SHEET	
11-0	BEAVER	0051	B30	13 OF 68	
		0018	B35		
ROCHESTER TOWNSHIP					
REVISION NUMBER	REVISIONS			DATE	BY



INTERSECTION DETAIL
SR 0018 AND
VETERANS MEMORIAL BRIDGE



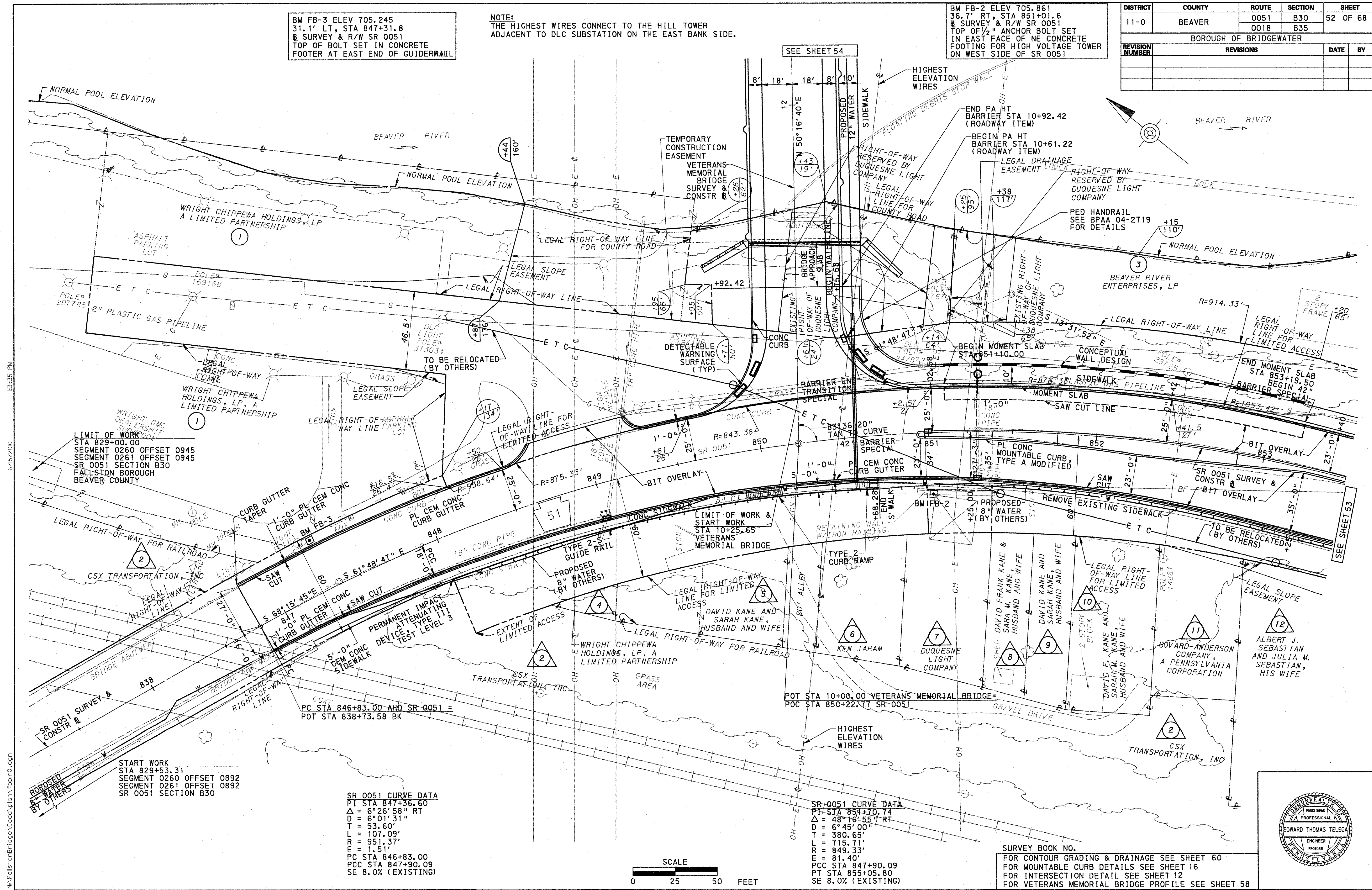
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BM FB-3 ELEV 705.245
31.1' LT, STA 847+31.8
SURVEY & R/W SR 0051
TOP OF BOLT SET IN CONCRETE
FOOTER AT EAST END OF GUIDERAIL

NOTE:
THE HIGHEST WIRES CONNECT TO THE HILL TOWER
ADJACENT TO DLC SUBSTATION ON THE EAST BANK SIDE.

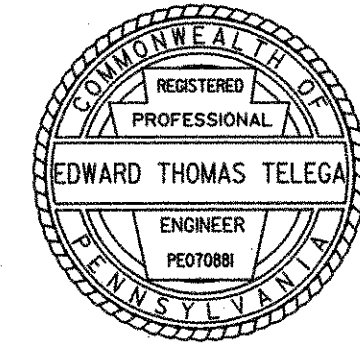
BM FB-2 ELEV 705.861
36.7' RT, STA 851+01.6
SURVEY & R/W SR 0051
TOP OF 1/2" ANCHOR BOLT SET
IN EAST FACE OF NE CONCRETE
FOOTING FOR HIGH VOLTAGE TOWER
ON WEST SIDE OF SR 0051

DISTRICT	COUNTY	ROUTE	SECTION	SHEET
11-0	BEAVER	0051	B30	52 OF 68
		0018	B35	
BOROUGH OF BRIDGEWATER				
REVISION NUMBER	REVISIONS			DATE BY



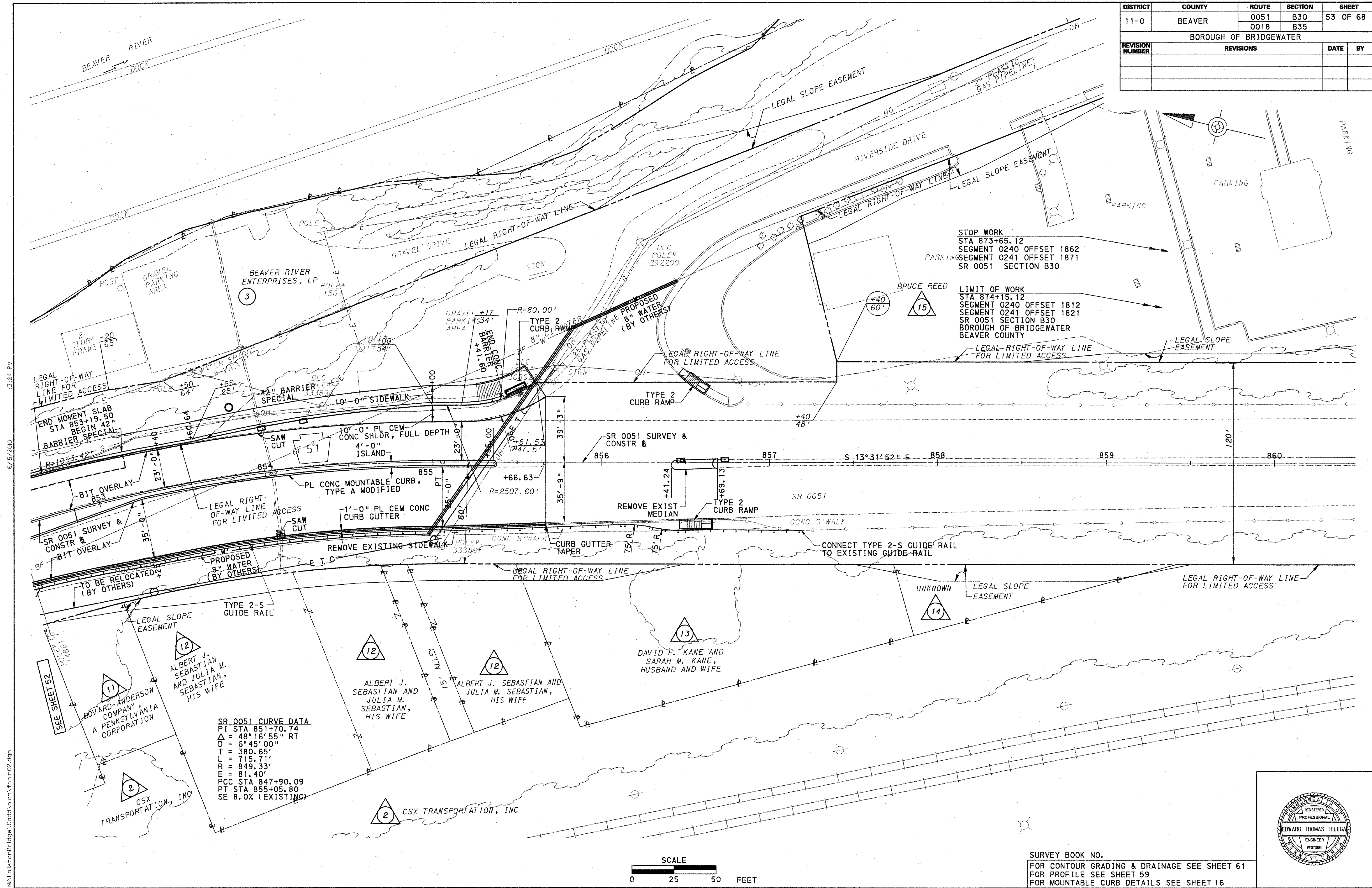
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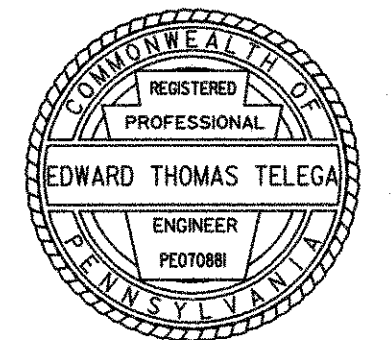
SURVEY BOOK NO.
FOR CONTOUR GRADING & DRAINAGE SEE SHEET 60
FOR MOUNTABLE CURB DETAILS SEE SHEET 16
FOR INTERSECTION DETAIL SEE SHEET 12
FOR VETERANS MEMORIAL BRIDGE PROFILE SEE SHEET 58

DISTRICT	COUNTY	ROUTE	SECTION	SHEET
11-0	BEAVER	0051	B30	53 OF 68
		0018	B35	
BOROUGH OF BRIDGEWATER				
REVISION NUMBER	REVISIONS			DATE BY



SR 0051 CURVE DATA
 PI STA 851+70.74
 $\Delta = 48^\circ 16' 55''$ RT
 $D = 6^\circ 45' 00''$
 $T = 380.65'$
 $L = 715.71'$
 $R = 849.33'$
 $M = 81.40'$
 PCC STA 847+90.09
 PT STA 855+05.80
 SE 8.0% (EXISTING)

SURVEY BOOK NO.
 FOR CONTOUR GRADING & DRAINAGE SEE SHEET 61
 FOR PROFILE SEE SHEET 59
 FOR MOUNTABLE CURB DETAILS SEE SHEET 16



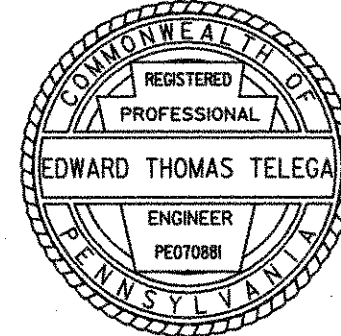
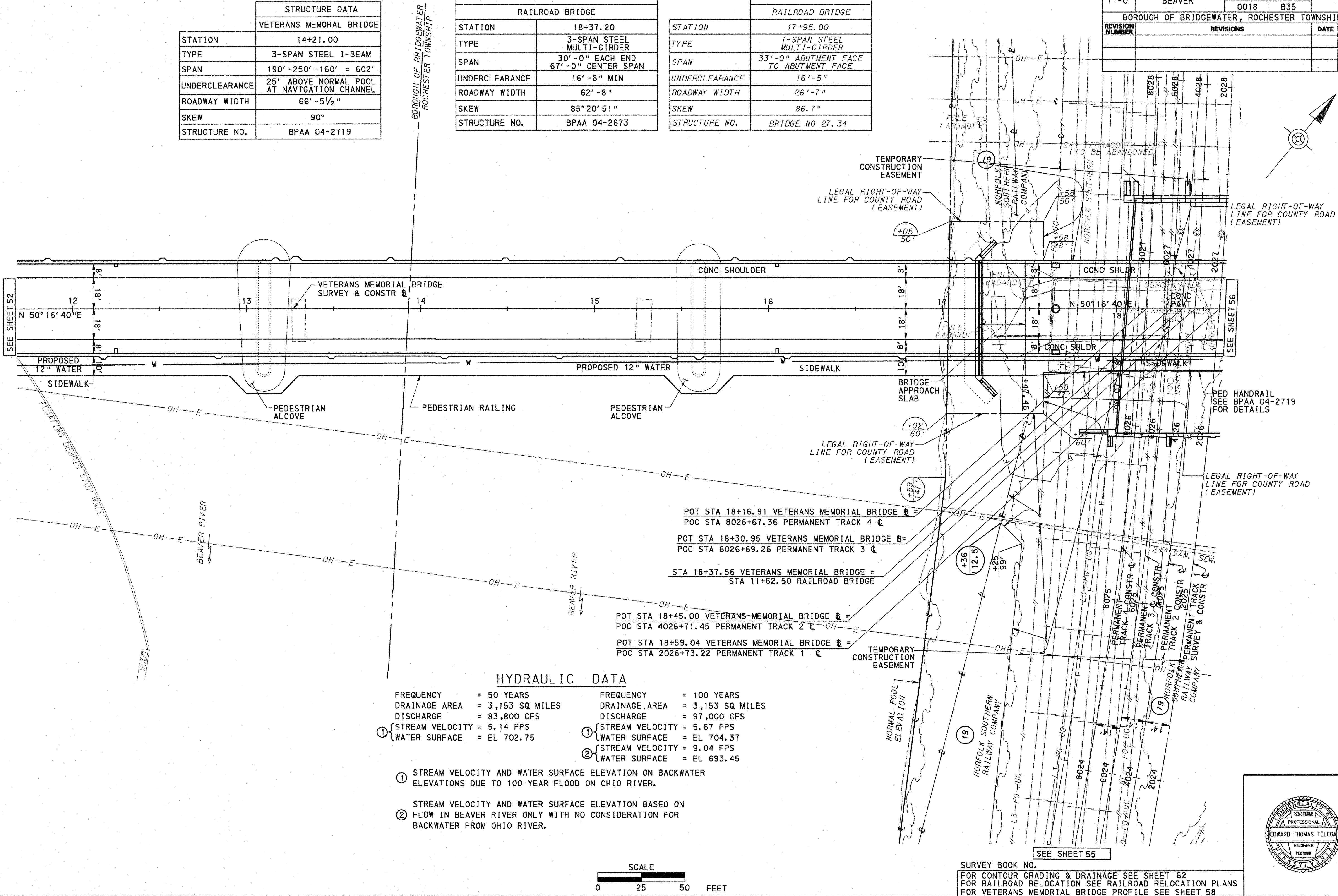
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STRUCTURE DATA	
VETERANS MEMORIAL BRIDGE	
STATION	14+21.00
TYPE	3-SPAN STEEL I-BEAM
SPAN	190'-250'-160' = 602'
UNDERCLEARANCE	25' ABOVE NORMAL POOL AT NAVIGATION CHANNEL
ROADWAY WIDTH	66'-5 1/2"
SKEW	90°
STRUCTURE NO.	BPAA 04-2719

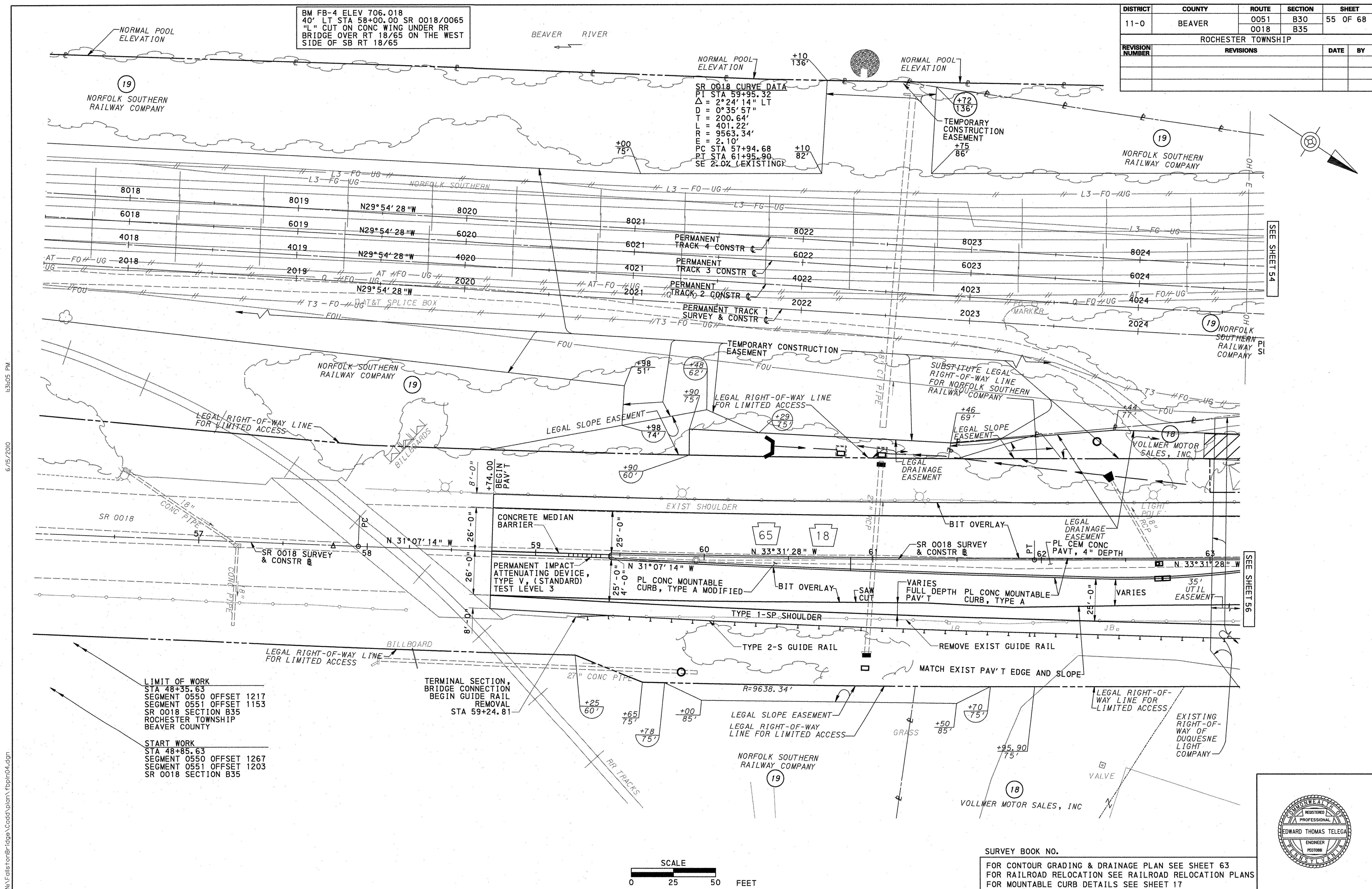
PROPOSED STRUCTURE DATA	
RAILROAD BRIDGE	
STATION	18+37.20
TYPE	3-SPAN STEEL MULTI-GIRDER
SPAN	30'-0" EACH END 67'-0" CENTER SPAN
UNDERCLEARANCE	16'-6" MIN
ROADWAY WIDTH	62'-8"
SKEW	85°20'51"
STRUCTURE NO.	BPAA 04-2673

STRUCTURE DATA	
RAILROAD BRIDGE	
STATION	17+95.00
TYPE	1-SPAN STEEL MULTI-GIRDER
SPAN	33'-0" ABUTMENT FACE TO ABUTMENT FACE
UNDERCLEARANCE	16'-5"
ROADWAY WIDTH	26'-7"
SKEW	86.7°
STRUCTURE NO.	BRIDGE NO 27.34

DISTRICT	COUNTY	ROUTE	SECTION	SHEET
11-0	BEAVER	0051 0018	B30 B35	54 OF 68
BOROUGH OF BRIDGEWATER, ROCHESTER TOWNSHIP				
REVISION NUMBER	REVISIONS			DATE BY



DISTRICT	COUNTY	ROUTE	SECTION	SHEET	
11-0	BEAVER	0051	B30	55 OF 68	
		0018	B35		
ROCHESTER TOWNSHIP					
REVISION NUMBER	REVISIONS			DATE	BY



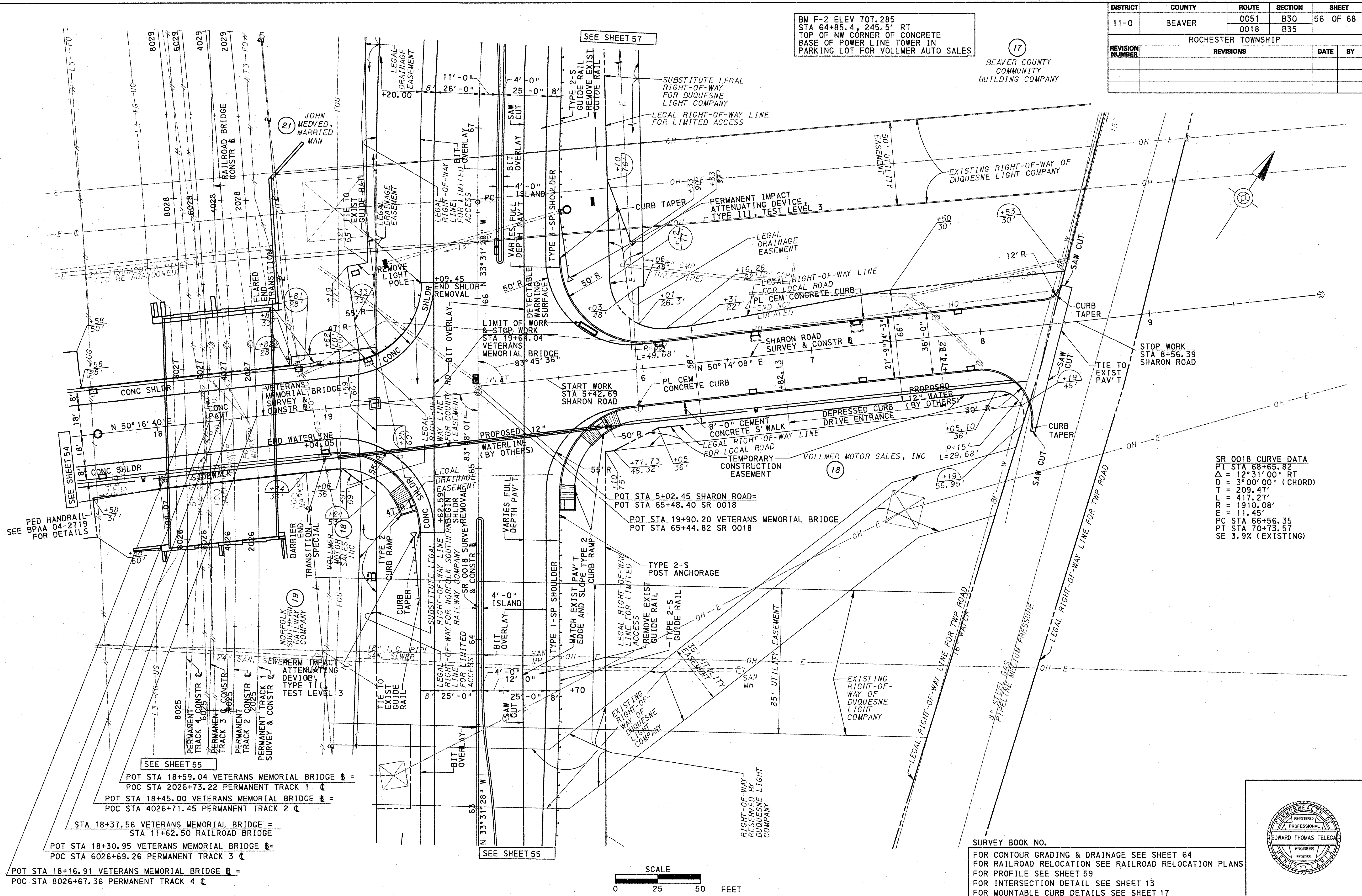
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COMMONWEALTH OF PENNSYLVANIA
REGISTERED PROFESSIONAL
EDWARD THOMAS TELEGA
ENGINEER
PE070881

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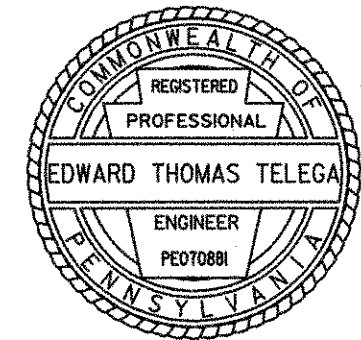
BM F-2 ELEV 707.285
STA 64+85.4, 245.5' RT
TOP OF NW CORNER OF CONCRETE
BASE OF POWER LINE TOWER IN
PARKING LOT FOR VOLLMER AUTO SALES

17
BEAVER COUNTY
COMMUNITY
BUILDING COMPANY

DISTRICT	COUNTY	ROUTE	SECTION	SHEET
11-0	BEAVER	0051	B30	56 OF 68
		0018	B35	
ROCHESTER TOWNSHIP				
REVISION NUMBER	REVISIONS	DATE	BY	

SR 0018 CURVE DATA
PI STA 68+65.82
Δ = 12°31'00" RT
D = 3°00'00" (CHORD)
T = 209.47'
L = 417.27'
R = 1910.08'
E = 11.45'
PC STA 66+56.35
PT STA 70+73.57
SE 3.9% (EXISTING)

SURVEY BOOK NO.
FOR CONTOUR GRADING & DRAINAGE SEE SHEET 64
FOR RAILROAD RELOCATION SEE RAILROAD RELOCATION PLANS
FOR PROFILE SEE SHEET 59
FOR INTERSECTION DETAIL SEE SHEET 13
FOR MOUNTABLE CURB DETAILS SEE SHEET 17



DISTRICT	COUNTY	ROUTE	SECTION	SHEET
11-0	BEAVER	0051	B30	57 OF 68
		0018	B35	
ROCHESTER TOWNSHIP				
REVISION NUMBER	REVISIONS	DATE	BY	

BM FB-5 ELEV 701.887
STA 74+57.6, 38.5' RT
"L" CUT IN SW CORNER OF CONCRETE
BASE OF PENNDOT ELECTRIC BOX
ON EAST SIDE OF RT 18/65

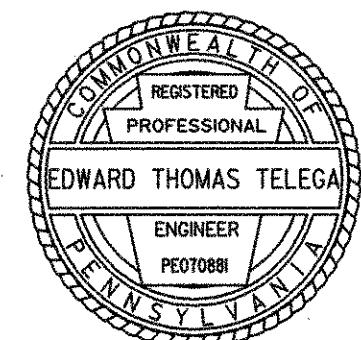
STOP WORK
STA 80+66.89
SEGMENT 0560 OFFSET 1516
SEGMENT 0561 OFFSET 1516
SR 0018 SECTION B35

LIMIT OF WORK
STA 81+16.89
SEGMENT 0560 OFFSET 1566
SEGMENT 0561 OFFSET 1566
SR 0018 SECTION B35
ROCHESTER TOWNSHIP
BEAVER COUNTY

SR 0018 CURVE DATA
PI STA 68+65.82
 $\Delta = 12^\circ 31' 00''$ RT
D = 3°00'00" (CHORD)
T = 209.47'
L = 417.27'
R = 1910.08'
E = 11.45'
PC STA 66+56.35
PT STA 70+73.57
SE 3.9% (EXISTING)

SCALE
0 25 50 FEET

SURVEY BOOK NO.
FOR CONTOUR GRADING AND DRAINAGE PLAN SEE SHEET 65
FOR RAILROAD RELOCATION SEE RAILROAD RELOCATION PLANS
FOR MOUNTABLE CURB DETAILS SEE SHEET 17

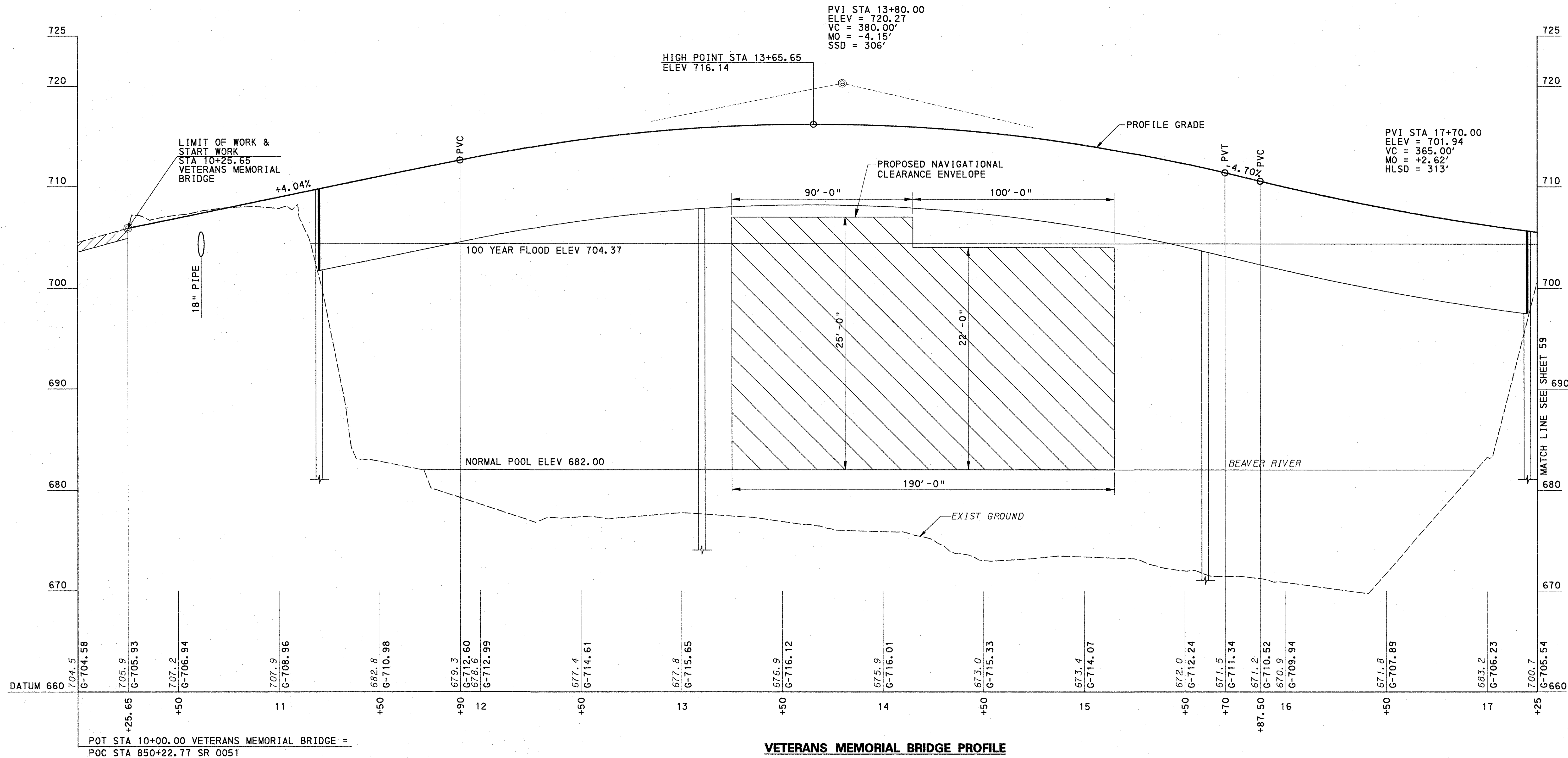


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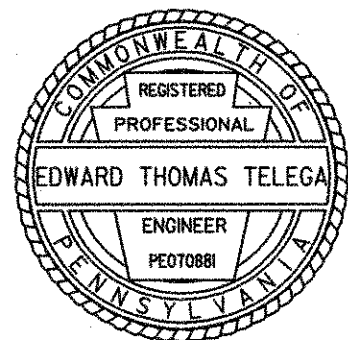
DISTRICT	COUNTY	ROUTE	SECTION	SHEET	
11-0	BEAVER	0051	B30	58 OF 68	
		0018	B35		
BOROUGH OF BRIDGEWATER, ROCHESTER TOWNSHIP					
REVISION NUMBER	REVISIONS			DATE	BY



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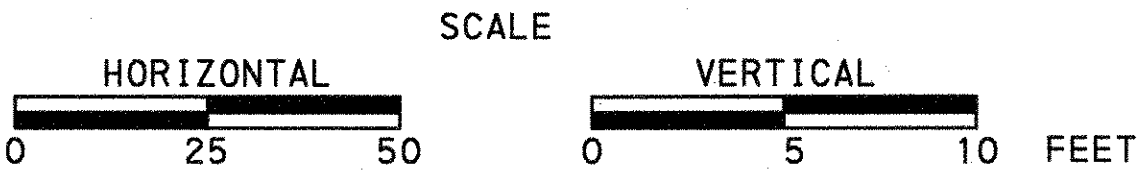
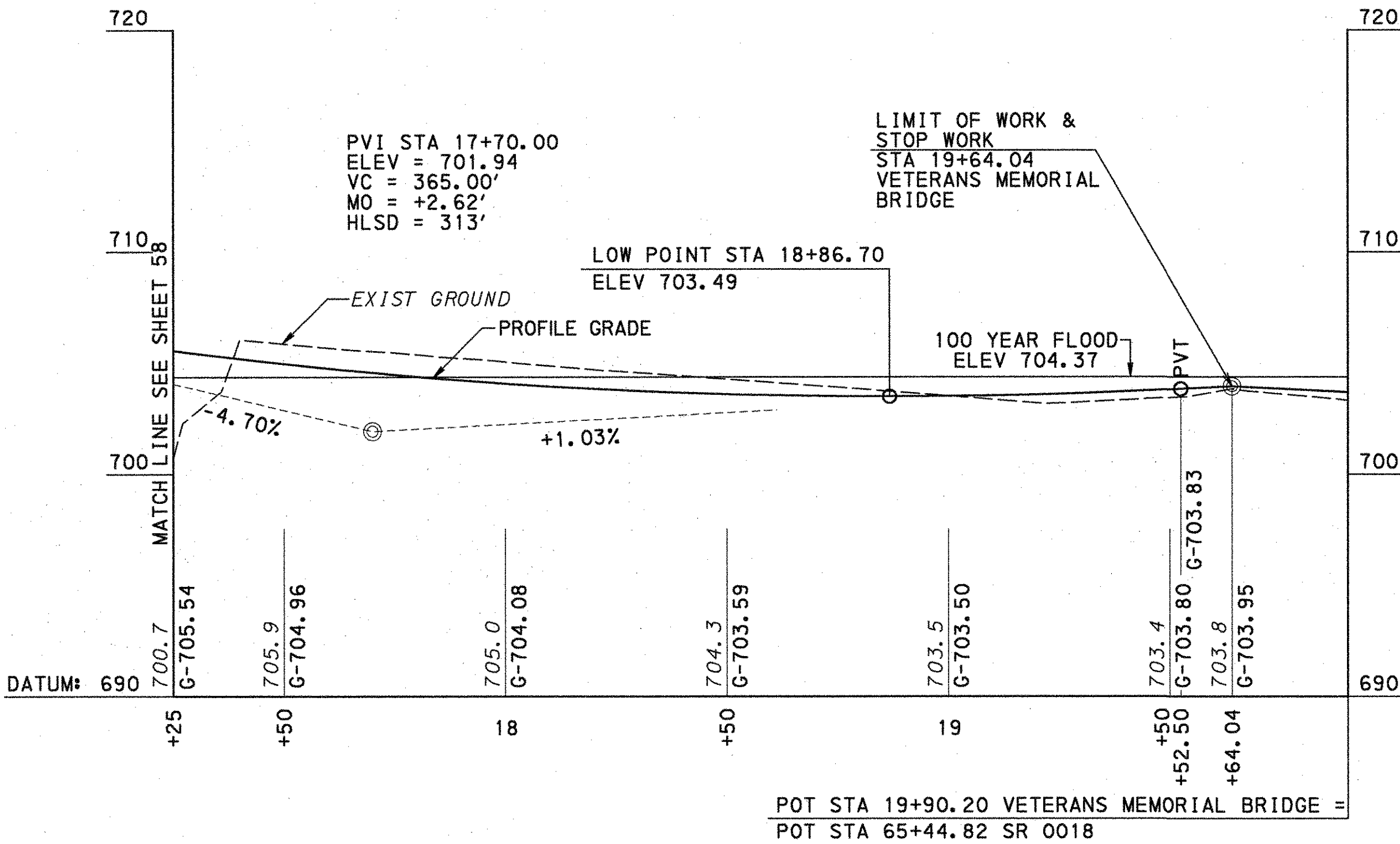
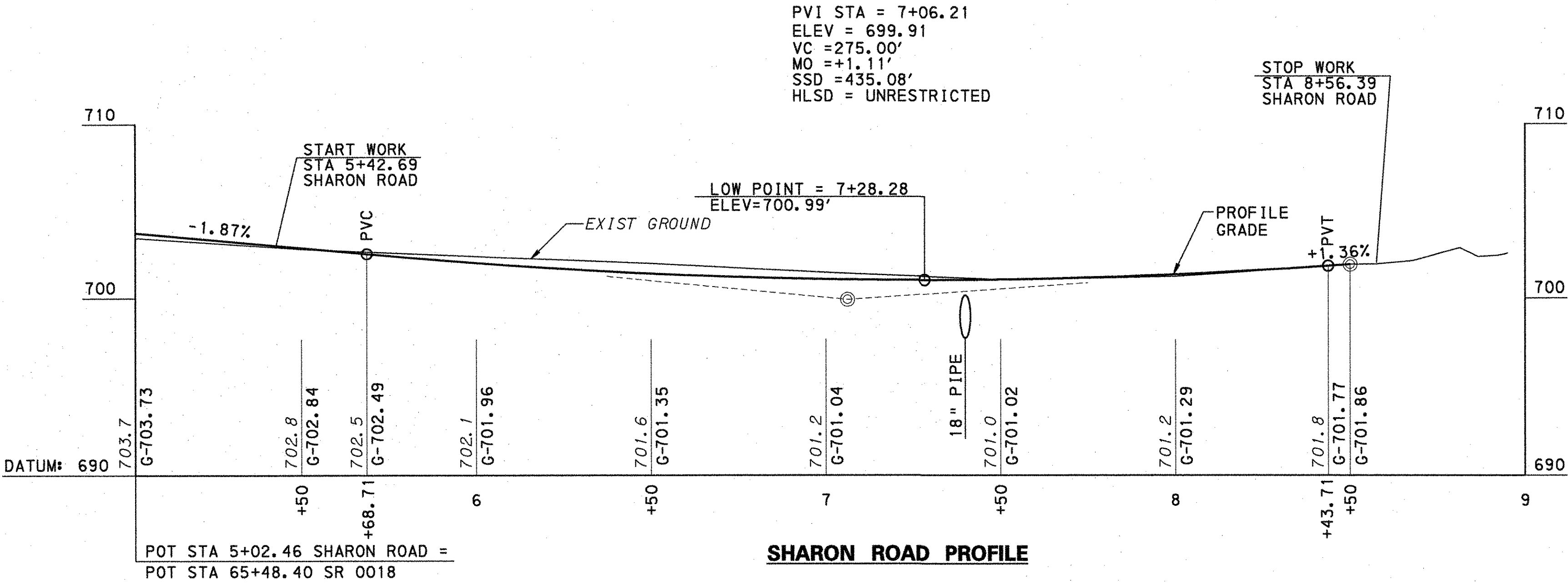
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SURVEY BOOK NO.
FOR PLAN SEE SHEETS 52 & 54

DISTRICT	COUNTY	ROUTE	SECTION	SHEET	
11-0	BEAVER	0051	B30	59 OF 68	
		0018	B35		
ROCHESTER TOWNSHIP					
REVISION/NUMBER	REVISIONS			DATE	BY



SURVEY BOOK NO.
FOR PLAN SEE SHEETS 54 & 56

